PREET PATEL

San Francisco Bay Area, CA · patelpb96@gmail.com · 630-715-6148 · LinkedIn · GitHub · patelpb.com

Experience

U.S. Patent & Trademark Office · *Patent Examiner, Photonics & Optical Systems* *Oct 2024 – Present*

- *Technical Architecture & Prior-Art Analysis.* Evaluate complex optical circuitry, semiconductor, and photonics systems against prior-art literature and statutory compliance standards; authored **100+ technical office actions** under strict production quotas.

Mercor · *AI Evaluation & Model Quality Specialist* (*Contractor*) *Apr 2026 – Present*

- *Frontier Model Evaluation.* Execute high-volume human-in-the-loop evaluations and rubric-based audits of frontier LLM outputs for multi-step reasoning, factual grounding, and instruction adherence; selected for priority fast-turnaround batches based on high inter-annotator agreement and diagnostic precision in surfacing model failure modes.

GlobalLogic Inc. (Google RMI Team) · *LLM Quality & Safety Analyst* *Jul 2024 – Oct 2024*

- *Adversarial & Edge-Case Auditing.* Conducted structured evaluations of Gemini and Magi search/reasoning outputs against safety and truthfulness guidelines, categorizing hallucination patterns, logic breakdowns, and data anomalies.

Education

University of California, Davis · *M.S., Physics* *2020 – 2023*
University of Michigan, Ann Arbor · *B.S., Physics; B.S., Astronomy & Astrophysics; Minor in Statistics* *2015 – 2019*

Research Experience

Type Ia Supernova Delay-Time Distribution Inference · *Independent Research* (*FIRE Collaboration*) *2026*

- Built a Bayesian inference pipeline comparing ten delay-time-distribution families (power-law through skewed-peak models) against Milky Way stellar abundance data from APOGEE DR17, using the element-tracer method on FIRE-2 cosmological simulations.
- Wrote a factorized yield integrator approximately 25× faster than the original numerical integration while agreeing to about one part in a million.
- Maintains a fork of the FIRE analysis code (GizmoElementTracers) with MCMC-based parameter estimation over stellar [Mg/Fe]–[Fe/H] distributions.

Elemental Abundances of Simulated Low-Mass Galaxies · *UC Davis, Computational Astrophysics* *Mar 2021 – Sep 2023*

- Engineered reproducible Python and HDF5 data pipelines for >**1 PB** FIRE-2 cosmological simulation datasets on national supercomputing clusters, cutting runtime by **50%+** via vectorization, redundancy removal, and parallelization.
- Identified elemental abundance signatures of bursty star formation and satellite accretion in present-day abundance distributions of low-mass galaxies.
- Built likelihood-based parameter estimation workflows (MCMC code) and gradient-descent models on noisy, high-dimensional (20+ feature) datasets to isolate statistically significant signals and quantify confidence bounds.
- Led end-to-end research from pipeline design to first-author publication in *MNRAS*, 512(4), 5671, collaborating across a 300+ scientist consortium with shared version-controlled software.

Projects

Rust Data Workbench · *Local-First CSV Exploration & Classical ML* *Oct 2026*

- Scoped a native Rust (egui, Polars, SmartCore candidates) alternative to notebook-stack data work: CSV import, data profiling, reversible cleaning/subsetting, interactive visualization, and supervised classification/regression with reproducible splits; local-only, no telemetry, source CSV never overwritten.
- Test-first build under a supervised multi-model pipeline: an 8-phase plan with a separate author and reviewer per slice, and a data contract plus an 85-file red-test bundle (27 data-import behavior tests written to fail against stubs) refined through nine independently reviewed revisions. No app code applied yet; the latest patch gate is held on build-script sandboxing fixes before implementation begins.
- *Planned.* CSV/profile core, transform engine, desktop shell, charts, ML pipeline, then end-to-end smoke tests and independent UX/code review.

Agent Workbench · *Tauri · Rust · Svelte · mouse-friendly tmux for AI-agent supervision* *Oct 2026*

- Built a Windows desktop workspace on a persistent terminal service (broker daemon plus one host process per terminal): terminals survive closing the app, restarting the broker, and rebuilds, and are re-adopted on restart. Implements tmux's default prefix keys (split tree to 64 panes, repeat-resize, zoom, copy mode, scrollback search, synchronized input, output logging), plus a Canvas of floating terminals and a scriptable `wb` CLI.

- Observer and Inspector read Claude Code, Pi, and Codex transcripts to show every agent session's state, tools, usage, and model, with an agent map; agents may only propose commands, and type into a terminal under time-limited human grants (dormant when no human is attached).
- Hardened the service (current-user-only DACL, HMAC challenge with DPAPI-protected secret, elevated clients refused, strict clippy gate, Pi/Argon adversarial review with all findings fixed); an 8-hour, 46k-round soak test traced memory growth to a per-connection security-descriptor leak, fixed and re-verified; an end-to-end test proves detach, broker restart, re-attach, grants, and full process teardown. Role-split governance: Claude designs/reviews, DeepSeek makes volume edits, guard.py gates destructive actions (allow/deny/escalate), with per-task evidence in a versioned handoff tree.
- *Planned.* Real sign-in and credential handling for other users, packaging and signing, and a free, bring-your-own-key release.

R2S2 · *Godot 4 .NET (C#) Skating/Movement Game* *Oct 2026*

- Rebuilt character movement as a JSRF-style surface state machine: a probe-validated surface frame (seven point probes, facet-spike-filtered normals) replaces rigid-body ground adhesion; tangent-plane RideMotion holds a 49° pipe shell with no minimum speed. Probe-driven movement is now the default, with the legacy controller kept selectable for comparison.
- Decoupled animation from physics with an analytic two-bone leg solve; deterministic `verify.sh` gate runs the 59-check friction suite (30/60/120 Hz), BotQC (50 checks), and the 83-check legacy suite green before any merge.
- Root-caused and fixed a lateral-friction bug (the scrub projected onto an axis built from the velocity it measured, so it was identically zero) and a steep-pipe sideways-landing bug (stale takeoff normal rejected the floor); added a route-clearance gate and rig-clearance measurement so fixes are tied to actual mesh clearance, not origin position.
- *Planned.* Human playtest of landing and carving feel, rails as curve traversal and wall ride as plane traversal at parity with legacy, then flip default and retire legacy; design specs written for graffiti/notoriety, equipment and skills, death/loss, and MMO networking and hosting.

Agent Report · *Multi-Agent LLM Newsroom* *Oct 2026*

- *Multi-Agent Architecture.* Designed and shipped a hierarchical multi-agent system that researches, writes, fact-checks, and publishes a daily news edition: a frontier-model (Claude Opus) editor-in-chief delegates to Sonnet senior editors and Haiku reporters through assign, review, revise, and publish loops run headlessly by a Python daemon.
- *Cost & Runtime Optimization.* Cut per-edition cost **46%** ($1.08 → $0.58) and paid web searches **53%** (55 → 26) via a shared research desk that caches/deduplicates sources, prioritizes free RSS feeds before paid search, and tiers model routing; full editions compile in 2–3 minutes.
- *Offline Eval Harness & Local Classifier (Laya).* Built an offline evaluation harness for a local relevance classifier screening search candidates before LLM ingestion (571 labeled candidates across 8 editions; 4 prompt templates compared via ROC-AUC and precision/recall cutoffs). The winning headline-only template (**AUC 0.96**) prunes **59%** of noise at 100% production recall.
- *GPU Inference Acceleration.* Migrated local classifier inference from CPU to CUDA (RTX 4060 Ti), cutting scoring latency ∼**17**× (406 → 23 ms/candidate); gated rollout behind shadow-mode logging to monitor drift.
- *Cloud Infrastructure & Guardrails.* Enforced a declarative source policy (≥2 independent sources for major claims, automated confidence labels for single-source signals). Deployed delivery over an encrypted AWS relay (**Lambda, CloudFront edge auth, S3, SSM**).

Publications

- S. Chughtai, S. Kamath, K. Patel, **P. Patel**, M.U. Ghani. "Predictors and contributors of burnout among hospitalists in central Michigan." *Journal of Healthcare Risk Management* (2026). PMID: 42307986
- **P. Patel**, S. Loebman, A. Wetzel, C.-A. Faucher-Giguère, K. El-Badry, J. Bailin. "Predictions for Complex Distributions of Stellar Elemental Abundances in Low-Mass Galaxies." *Monthly Notices of the Royal Astronomical Society*, 512(4), 5671 (2022). doi:10.1093/mnras/stac834

Skills

AI & LLM Systems	Multi-Agent Orchestration, Anthropic API (Opus/Sonnet/Haiku), Offline Eval Harnesses, Prompt Engineering & Ablations, Local Classifier Inference, CUDA, Rubric & Guardrail Design
Cloud, Data & Infra	Python, R, SQL, AWS (Lambda, S3, CloudFront, SSM), C++, Polars, Pandas, NumPy, SciPy, HDF5, Rust, Tauri/Svelte, C#/Godot, Linux/Unix, Git
ML & Statistics	ROC-AUC / Precision-Recall Optimization, Classification, Statistical Inference, MCMC, Maximum Likelihood, Nonparametric Tests, Regression, Time-Series Analysis, Uncertainty Quantification